Decision Tree

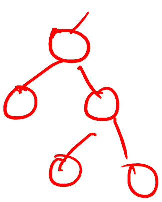
Wednesday, May 20, 2026 11:35 AM

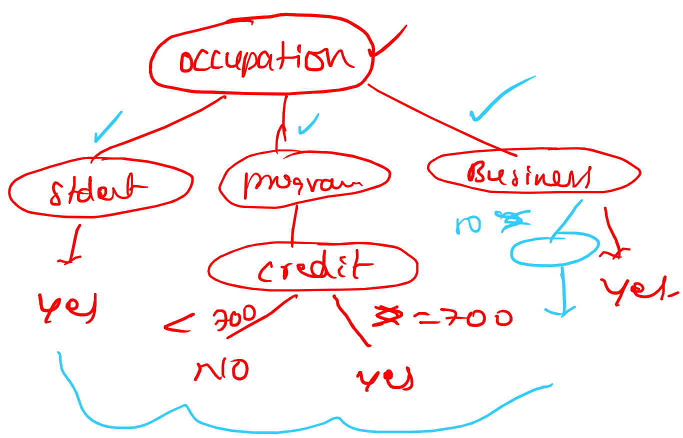
Decision Tree :

- It is also supervised ML algorithm
- Used for both classification & Reg.

Age	occupation	credit	purchase
25	student	600	yes
17	student	610	yes
33	programer	710	no
45	Business	800	yes
55	Business	850	yes

```
if occupation=="student"
    print("yes")
else:
    if credit >=780:
        print("yes")
    else:
        print("no")
```

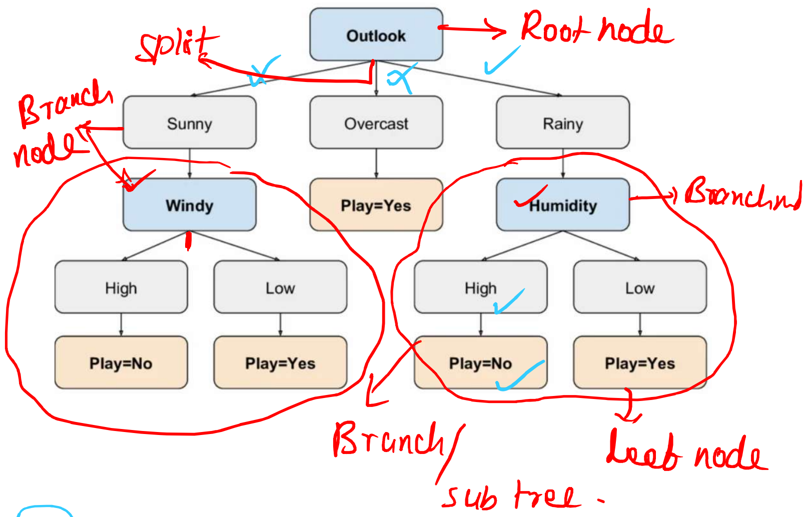




Bus, 80, 850 ?

1 2 3 4

Outlook	Temperature	Humidity	Windy	PlayTennis
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	True	No



Rainy, cool, high, True ?

1. How to select Root node?

2. How to select Branch node?

3. Pruning / cutting.

1. Entropy

- Need to calculate entropy for each column
- Entropy is nothing but uncernity / noisy in a data.

Entropy value = low (less noise)

Entropy value = high (uncertinity)

2. Information Gain.

- How much entropy can be reduced can be find using IG.

IG value = high (more Information)

IG value = low (less information).

1. How to select Root node?

0. Calculate entropy for Target column.

1. Calculate Entropy for each column.

2. Find the unique values in the column

3. Find weighted entropy for each unique value.

4. Find IG for each column

— select highest IG value (your root node)

5. Repeat the same process for each sub tree.

① 2 3 4

Outlook	Temperature	Humidity	Windy	PlayTennis
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	True	No

$$\text{Entropy} = \sum P_i \log(P_i)$$

$$= P_1 \log(P_1) + P_2 \log(P_2) + P_3 \log P_3 \ldots$$

$$IG = \text{Entropy (parent)} - \text{Weighted Entropy}.$$

step 1: Calculate Entropy for target

Total = 14.

yes = 9, no = 5.

p(yes) = 9/14, p(no) = 5/14.

$$\text{Entropy(parent)} = P_{yes} \cdot \log(P_{yes}) + P_{no} \cdot \log(P_{no})$$

$$= 9/14 \cdot \log[9/14) + 5/14 \cdot \log(5/14)$$

$$= \log(9/14) = -0.64, \ \log(5/14) = -1.48.$$

$$= (-0.64 \times 9/14) + (-1.48 \times 5/14) = 0.94$$

2. Calculate Entroy for outlook.

outlook	yes	NO	Total
sunny	2	3	5
overcast	4	0	4
Rainy	3	2	5
			14.

1. $\text{Entropy(sunny)} = 2/5 \log(2/5) + (3/5) \cdot \log(3/5)$

$$= 0.97$$

2. $\text{Entropy(overcast)} = 4/4 \cdot \log(4/4) + 0/4 \cdot \log(0/4)$

$$= 0 + 0 = 0.$$

3. $\text{Entropy(Rain)} = 3/5 \cdot \log(3/5) + 2/5 \cdot \log(2/5)$

$$= 0.97.$$

step 3: Find weighted entropy for Outlook:

Step 3: Find weighted entropy for Outlook:

$$\text{Entropy(outlook)} = \frac{5}{14} \times 0.97 + \frac{4}{14} \times 0 + \frac{5}{14} \times 0.97$$

$$= \underline{0.69}$$

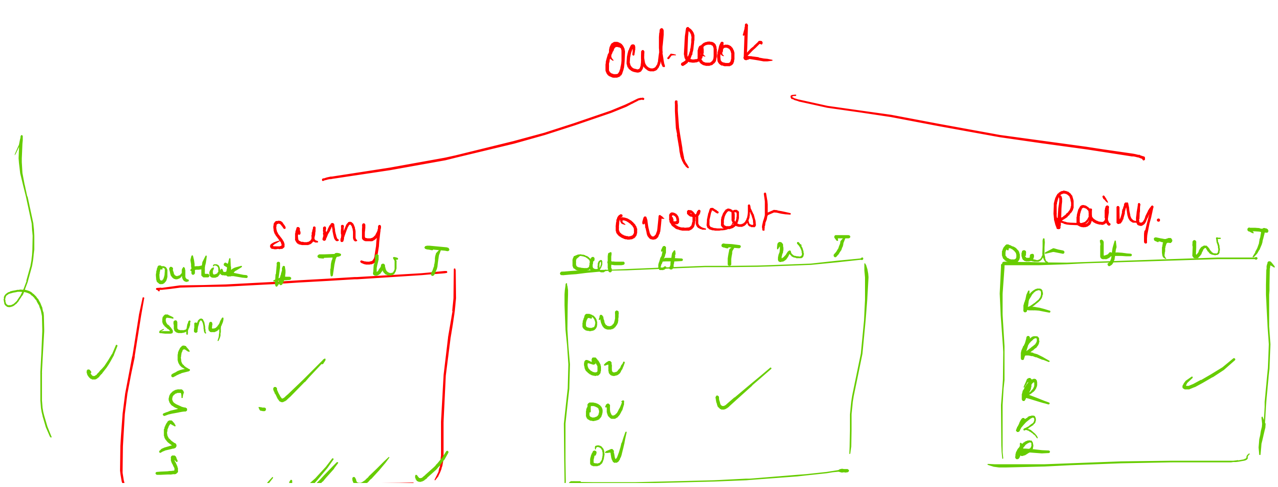
Step 4: Calculate IG for outlook.

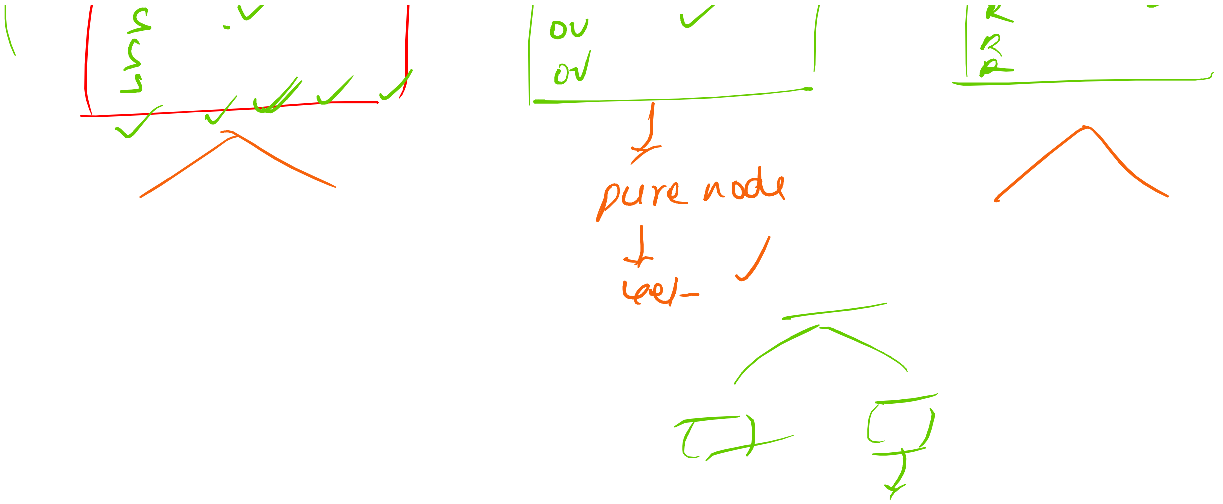
$$IG = \text{Entropy (parent)} - \text{Weighted Entropy}$$

$$= 0.94 - 0.69$$

$$= \boxed{0.25}$$

Feature	IG
Outlook	$\boxed{0.25}$
Humidity	0.151
Temprature	0.029
windy	0.048.





One more Technique to find Root node:

1. Gini

— It is uncertainty in data.

$$Gini = 1 - \sum P_i^2$$

$$= 1 - (P_1^2 + P_2^2 + P_3^2)$$

1. Find Gini for Target column

$$= 1 - \left((5/14)^2 + (9/14)^2 \right)$$

$$= 1 - \left(25/196 + 81/196 \right)$$

$$= 1 - 106/196 = \underline{0.459}.$$

2. Find Entropy for out-look column.

$$Gini(Sunny) = 0.48.$$

Gini(overcast) = 0.

Gini(Rainy) = 0.48.

3. Find the weighted Gini

$= (5/14) \times 0.48 + (4/14) \times 0 + (5/14) \times 0.48$

$= \boxed{0.34}$

4 Find IG (Gini Gain)

= Gini(parent) — W.Gini

$= 0.459 - 0.34 = \boxed{0.119}$

Features	Gini
outlook	0.34
Temp	0.44
Windy	0.43
Humidity	0.37

```
plt.scatter(x = cost, y= spicy, c=y, s= 100,
```

Advantages of DT:

— 1. It can used for both class & Reg.

2. Human friendly.

3. Can handle missing values & outlier.

4. No need of scaling, encoding.

5. Can give important feature.

Age

35

25

15

200

Age > 80

Flow of DT formation:

Select entire dataset

↓

Compute entropy on each feature.

↓

Compute weighted entropy

↓

Compute IG on each feature.

↓

select best feature (Root node)

↓

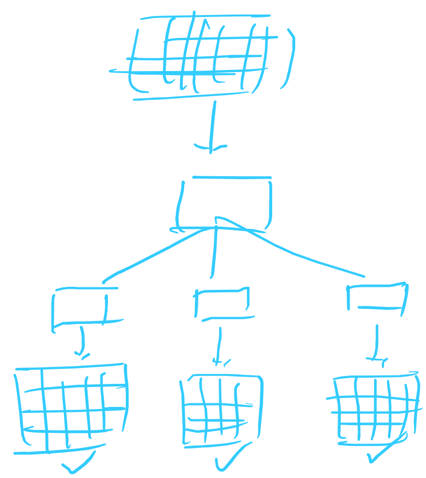
split the feature.

↓

Each split having sub dataset

↓

Repeat the same procedure for each branch.



1. Entropy

$$= \sum P_i \log P_i$$

$$= P_1 \log P_1 + P_2 \log P_2 + P_3 \log P_3$$

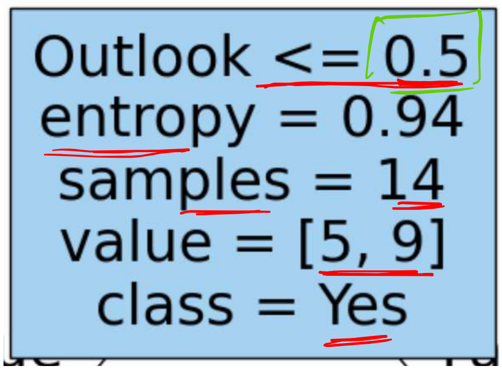
Weighted Entropy = wt × en(u1) + wt × en(u2) + wt × en(u3)

2. IG = Entrop(parent) − weighted Entropy

= − ✓

3. Gini = $1 - \sum P_i^2$

$= 1 - (P_1^2 + P_2^2 + P_3^2)$



$\boxed{\text{Sunny}}$	0 ✓	0.5	
overcast	1 ✓	$\boxed{1.5}$	
Rain	2 ✓		

out <= 1

Age

30 age <= 45.

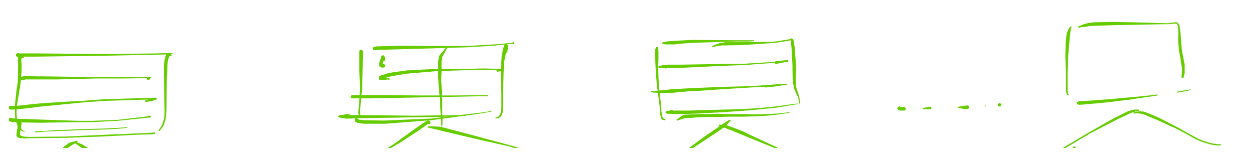
60

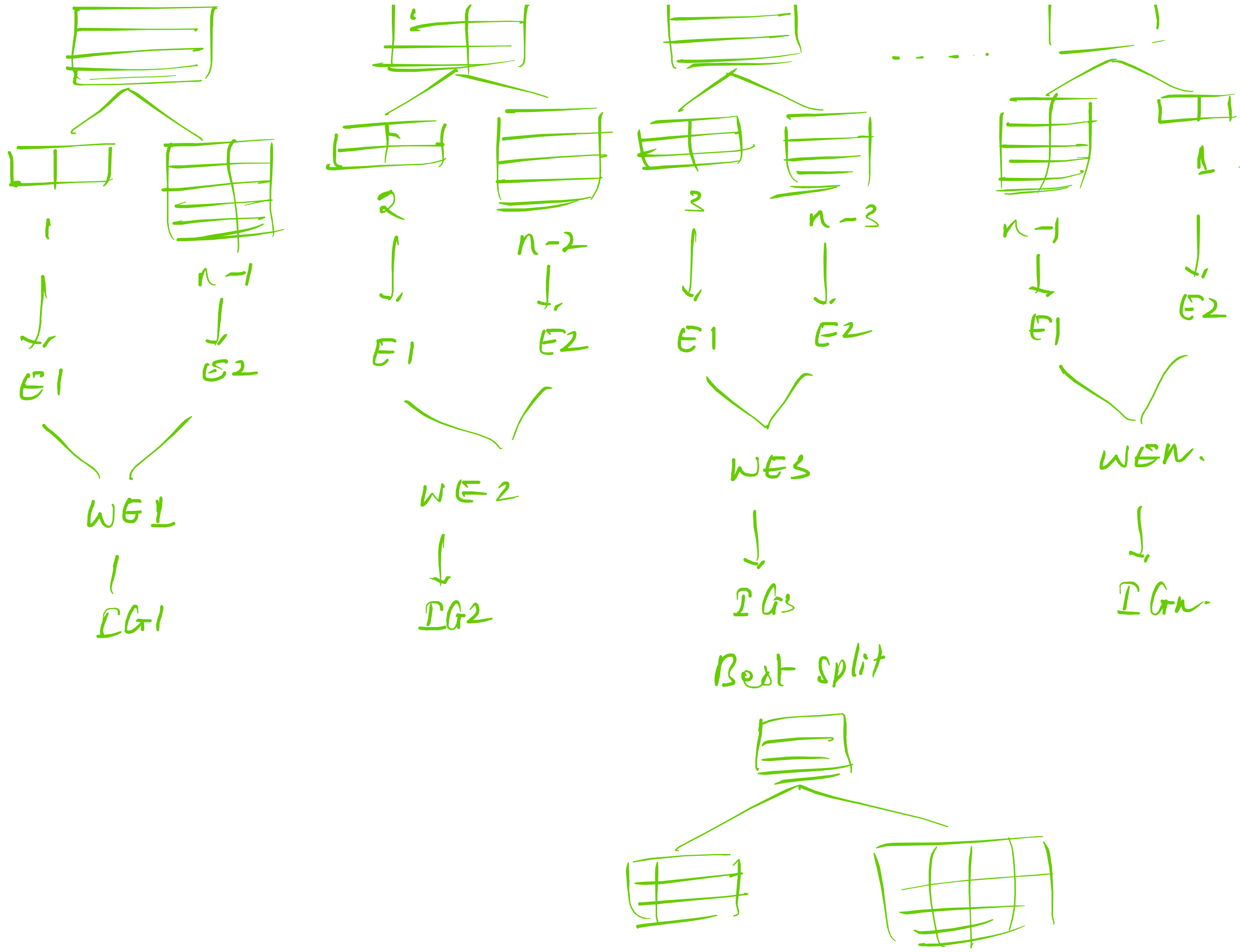
DT As Greedy Algorithm ✓

Handling Continuous data :

Age	purchase
17	yes
33	yes
25	no
23	no
55	yes
43	no.

1. Sort the feature.
2. select first row.





Different Algorithms inside DT:

1. ID3 (Iterative Dichotomiser 3):

— It can work for only classification problems.

— Uses Entropy internally.

— can not handle missing data

— can not handle continuous data well.

2. C4.5

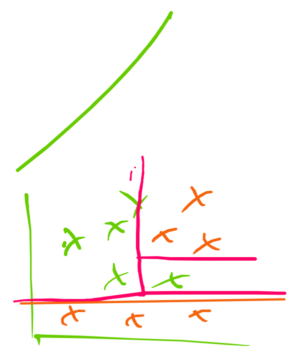
— It is an advanced version of ID3.

- It can handle continuous data
- Used technique call Gini ratio.

3. CART (Classification and Regression Tree)

- This is the popular tech used
- Even sklearn uses this CART as default.
- Uses Gini internally.
- can handle both classification & Reg.

Advantages of DT:



1. works for both classi & Reg
2. No need of any scaling & Encoding.
3. Can handle non-linear data
4. Can handle missing value.

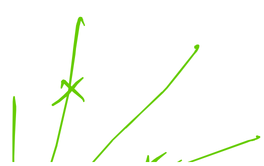
   1. Ignore that row.
   2. Apply some saming with probabilistically.
   3. Surrogate spliting.

   Age   Incoming

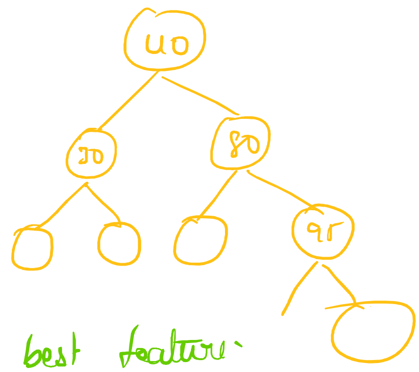
   -

5. Can handle Outliers.

Age



Age
30
45
23
55
200



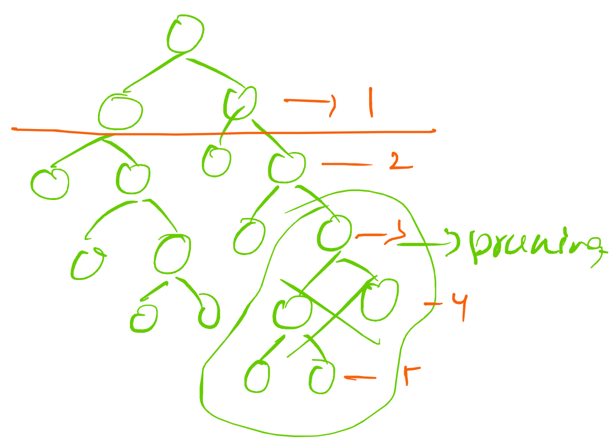
6. It can identify / suggest best feature.

Issues with DT :

1. Overfitting ✓
2. Underfitting

max_depth = 1

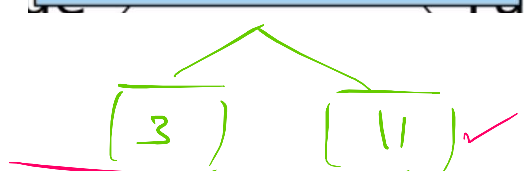
max_depth = None.



1. Features in DT :

1. splitter < best / random ✓
2. max_depth = None 5
3. min_sample_split = ~~2~~ 10
4. min_sample_leaf = ~~1~~ 3
5. max_features = None 5

Outlook <= 0.5
entropy = 0.94
samples = 14
value = [5, 9]
class = Yes



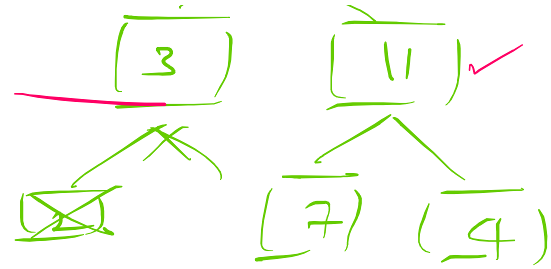
10 col

$x_1, x_2, x_3 \dots x_{10}$

$x_3, x_5, x_9, x_6, x_{10}$

x_9

(x_9)



Why we are calling DT as Greedy Algorithm.

- In DT, the split will happen on local ~~tr~~ node, which assume the outcome will the best global optimal solution.

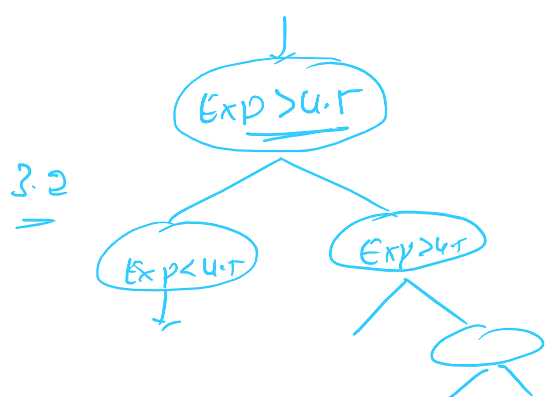
    - Entropy / Gini
    - IG

outlook → highest → IG.

- Going with best IG.

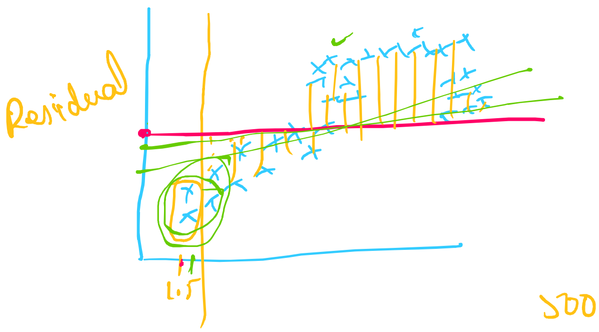
Regression:

Experience	Salary
1	20k
2	40k
3	70k
4	
5	90k
6	1L
7	1.5L

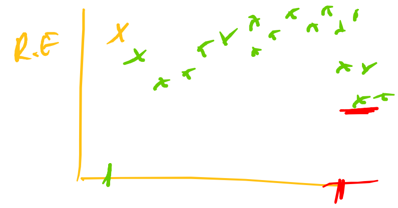


6
7

1L
1.5L



$MSE = \frac{1}{n}\sum(y-\hat{y})^2$




```
df["Quality"]= df["quality"].map(lambda x: "Good" if x>5 else "Bad")
```

dtreeviz

$\text{Bais} \propto \frac{1}{\text{var}}$

Underfitting	Overfitting	
low var	high var	low var
high bias	low bias.	low bias